PROBLEM-BASED LEARNING MODULE

Fuzzy Logic Systems

A Worked Case Study: Designing a Smart Irrigation System

Soil Moisture | Temperature | Rainfall | → **Irrigation Duration**

Inputs (3) • Output (1) • Mamdani Fuzzy Inference

Course: Fuzzy Logic & Genetic Algorithms

THE PROBLEM

Why Does a Sprinkler Need Fuzzy Logic?

Scenario

A farm wants to automate its sprinklers. A simple rule like “if soil moisture < 30%, turn on water for 20 minutes” ignores that a hot, dry, rain-free day needs far more water than a mildly dry day before a shower. Human experts reason in vague terms — “fairly dry,” “quite hot,” “light rain” — not sharp thresholds. Fuzzy logic lets a controller reason the same way.

Design Task

Build a fuzzy inference system that reads Soil Moisture, Temperature and Rainfall, and outputs an Irrigation Duration — without a single hard-coded threshold.

Learning Objectives

1. Differentiate crisp sets from fuzzy sets
2. Design membership functions for real variables
3. Fuzzify crisp sensor readings
4. Build a fuzzy IF–THEN rule base
5. Apply Mamdani inference (min / max)
6. Aggregate rule outputs and defuzzify (centroid)
7. Evaluate fuzzy logic vs. threshold control

Crisp Sets vs. Fuzzy Sets

Crisp (Classical) Set

Membership is binary: an element either fully belongs (1) or does not belong (0) at all.

```
"Soil is DRY"  →  Moisture < 30%  (else NOT dry)
```

29% ⟶ "Dry" (μ = 1)   31% ⟶ "Not dry" (μ = 0)

Problem: a 2% sensor difference flips the decision completely — an unrealistic, brittle boundary.

Fuzzy Set

Membership is a matter of degree: μ(x) ∈ [0, 1], defined by a membership function.

```
μ_Dry(x)  varies smoothly from 1 → 0
```

29% ⟶ "Dry" (μ = 0.38)   31% ⟶ "Dry" (μ = 0.34)

A reading can be "somewhat dry" AND "somewhat moist" at once — matching how experts actually describe conditions.

Architecture of a Fuzzy Inference System

Knowledge Base
(Expert Rules)

1

Fuzzification

Convert crisp sensor inputs into fuzzy membership degrees using membership functions.

2

Rule Base & Inference Engine

Apply expert IF–THEN rules; combine antecedents with min / max (Mamdani).

3

Aggregation

Combine all fired rule outputs into one overall fuzzy output set.

4

Defuzzification

Convert the fuzzy output set back into one crisp, actionable number.

Crisp Inputs
(Soil %, °C, mm)

Formal Definition of a Fuzzy Set

A fuzzy set A on a universe of discourse X is a set of ordered pairs:

$$A = \{ (x, \mu_{\square}(x)) \mid x \in X \}, \qquad \mu_{\square} : X \to [0, 1]$$

Universe of Discourse (X)

The full range of possible crisp values a variable can take — e.g. Soil Moisture ∈ [0, 100]%.

Membership Function $\mu_a(x)$

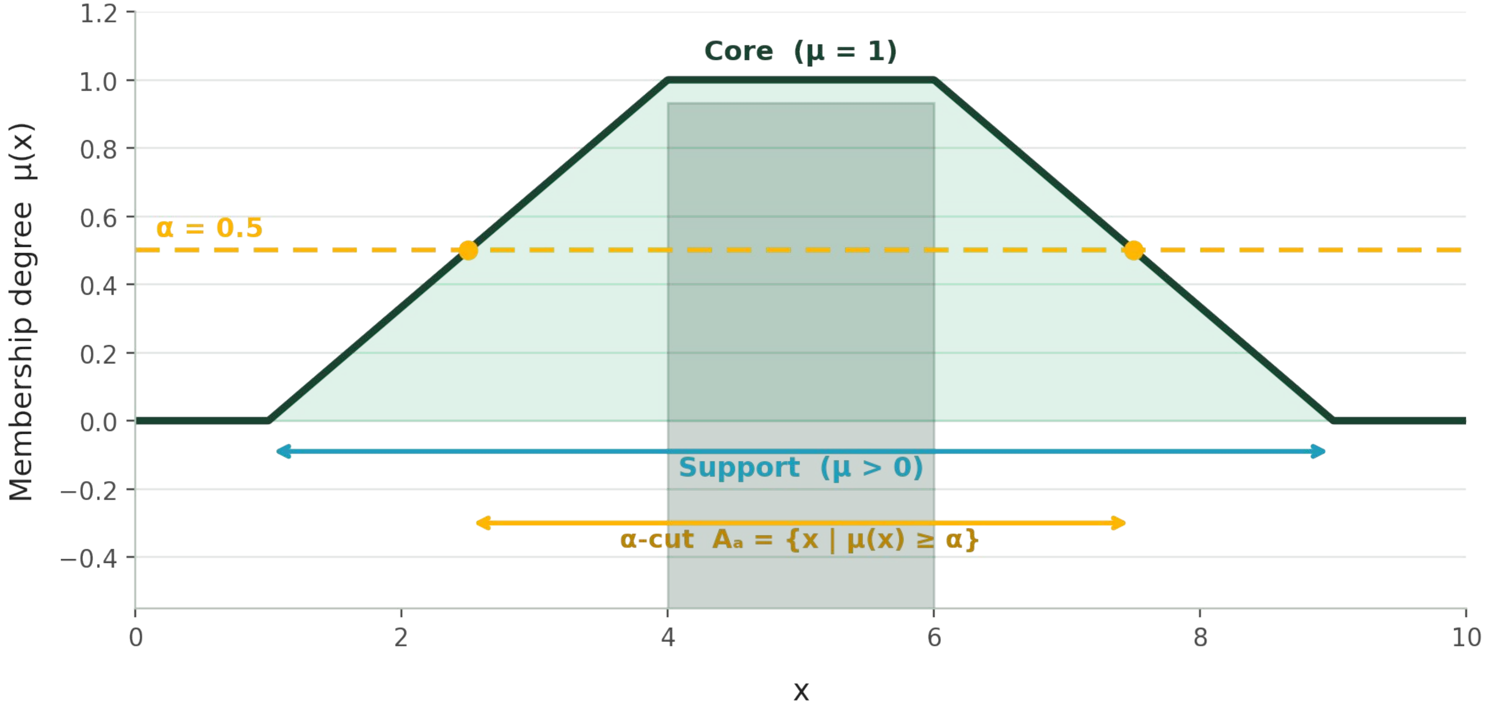
A curve that maps every x in X to a degree of belonging in [0,1] — not just {0, 1}.

Fuzzy Singleton

A single element with μ = 1 and everywhere else μ = 0 — a fuzzy set that reduces to one crisp point.

Discrete notation: $A = \Sigma_{\square}\ \mu_{\square}(x_{\square})/x_{\square}$ Continuous notation: $A = \int_{\square} \mu_{\square}(x)/x$ (the "/" here is notation, not division)

Key Properties of a Fuzzy Set



Support

{ x ∈ X | $\mu_a(x) > 0$ } — all x with any nonzero membership.

Core

{ x ∈ X | $\mu_a(x) = 1$ } — elements that fully belong.

Height

$\sup_x \mu_a(x)$ — the tallest point of the curve.

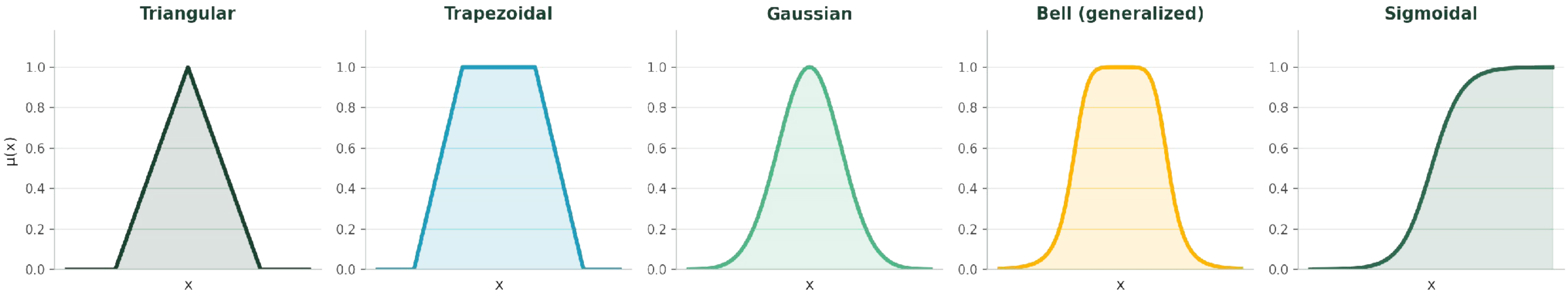
Normal Set

A fuzzy set with height = 1 (has a non-empty core).

α-cut (alpha-cut)

A_a = { x | $\mu_a(x) \geq \alpha$ } — the crisp set of elements at or above degree α.

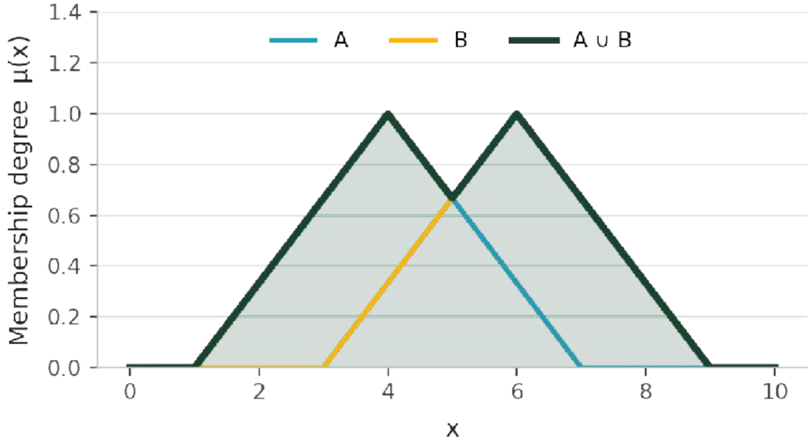
Common Types of Membership Functions



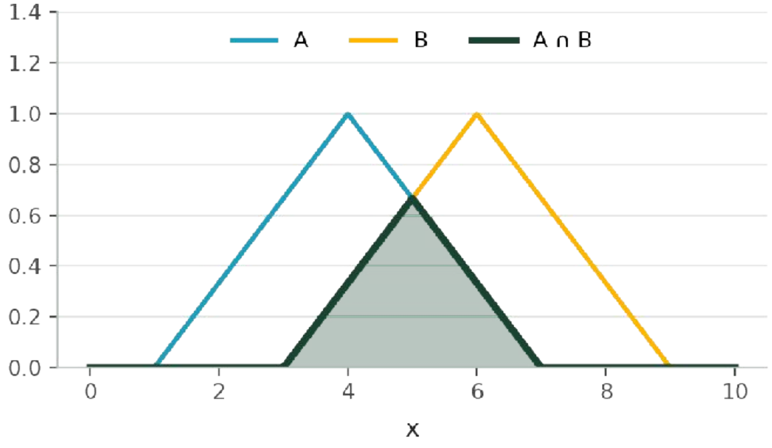
Shape	Parameters	Typical Use
Triangular	a, b, c (foot, peak, foot)	Simplest, cheapest to compute — our irrigation example
Trapezoidal	a, b, c, d (plateau between b, c)	A range of values equally “full” membership (e.g. “normal”)
Gaussian	mean, σ (spread)	Smooth, differentiable — common in neuro-fuzzy / ANFIS systems
Bell (generalized)	a, b, c (width, slope, center)	Adjustable smoothness independent of width
Sigmoidal	a (slope), c (crossover)	Open-ended concepts, e.g. “very hot” with no upper bound

Fuzzy Set Operations (Zadeh Operators)

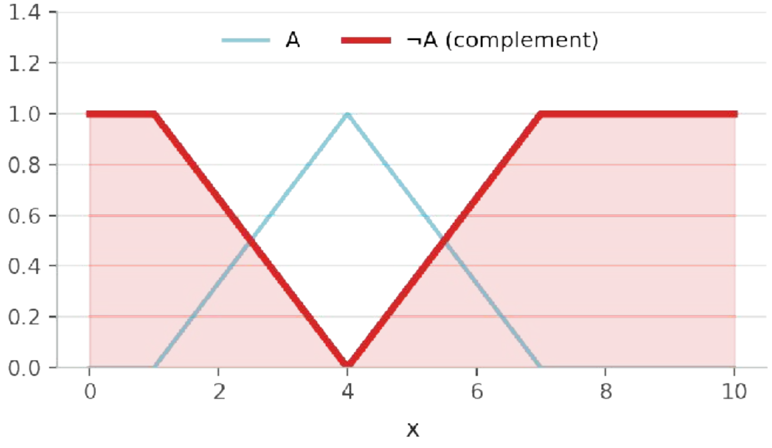
Union: $\mu_{a}\cup^{b}(x) = \max(\mu_{a}(x), \mu^{b}(x))$



Intersection: $\mu_{a}\cap^{b}(x) = \min(\mu_{a}(x), \mu^{b}(x))$



Complement: $\mu_{\neg a}(x) = 1 - \mu_{a}(x)$



Union (OR)

μ□∪□(x) = max(μ□(x), μ□(x))

Intersection (AND)

μ□∩□(x) = min(μ□(x), μ□(x))

Complement (NOT)

μ¬□(x) = 1 − μ□(x)

These are exactly the AND / OR operators the inference engine will use later to combine rule antecedents.

Generalized Operators: T-Norms & T-Conorms

min / max (Zadeh operators) are just one valid choice. Any function satisfying commutativity, associativity, monotonicity, and the right boundary conditions can serve as a generalized AND (t-norm) or OR (t-conorm).

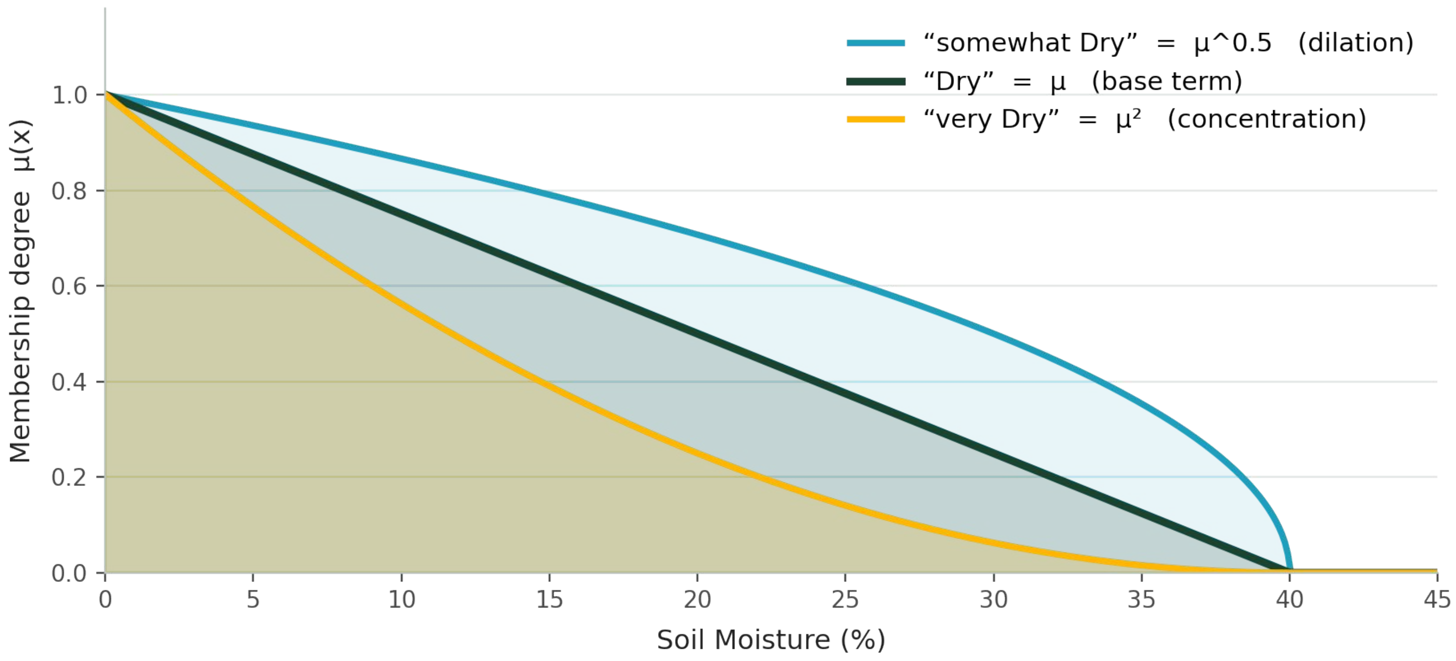
Family	T-Norm (Generalized AND)	T-Conorm (Generalized OR)	Notes
Zadeh (standard)	`min(a, b)`	`max(a, b)`	Used throughout this module — simple, fast, idempotent
Algebraic / Probabilistic	`a × b`	`a + b − a×b`	Smoother interaction; sensitive to every input
Bounded (Łukasiewicz)	`max(0, a+b−1)`	`min(1, a+b)`	"Strict" logic-style bounds; common in fuzzy control theory
Drastic	`b if a=1, a if b=1, else 0`	`b if a=0, a if b=0, else 1`	Extreme case — rarely used in practice, shows the outer limit

Why it matters

The choice of t-norm/t-conorm changes how sharply the system reacts to combined conditions. min/max is the standard teaching choice because it is intuitive and computationally cheap — exactly what this module uses.

Linguistic Variables & Linguistic Hedges

A linguistic variable takes words, not numbers, as its values — e.g. Soil Moisture ∈ {Dry, Moist, Wet} instead of a real number. Hedges modify a term's membership function to create new terms without new rules.



Concentration — "very A"

$\mu(x)^2$

Squares membership — narrows the set, makes it more restrictive.

Dilation — "somewhat A"

$\mu(x)^{0.5}$

Square-roots membership — widens the set, makes it more permissive.

Intensification — "indeed A"

`raises above/below 0.5 sharply`

Increases contrast around the crossover point.

Mamdani vs. Sugeno (TSK) Inference

Aspect	Mamdani	Sugeno (Takagi–Sugeno–Kang)
Rule consequent	Fuzzy set (“THEN Duration is Long”)	Function of inputs (“THEN y = p·soil + q·temp + r”)
Output before defuzz.	A full fuzzy set requiring aggregation	A crisp number per rule — no aggregation step needed
Defuzzification	Needed (centroid, bisector, …)	Simple weighted average of rule outputs
Interpretability	High — rules read like expert statements	Lower — consequents are mathematical, less intuitive
Computational cost	Higher (numerical integration)	Lower — well suited to optimization / adaptive tuning (ANFIS)
This module uses →	**Mamdani — chosen for teaching clarity**	—

Both are valid Fuzzy Inference Systems — the choice is a trade-off between interpretability (Mamdani) and computational efficiency / trainability (Sugeno).

CASE STUDY

Designing the Smart Irrigation FIS

Soil Moisture

% volumetric water content

Range: 0–100 %

Dry · Moist · Wet

Temperature

ambient air temperature

Range: 0–50 °C

Cold · Warm · Hot

Rainfall

recent / forecast rainfall

Range: 0–10 mm

None · Light · Heavy

→

Irrigation Duration

pump / sprinkler run-time

Range: 0–60 min

None · Short · Medium · Long

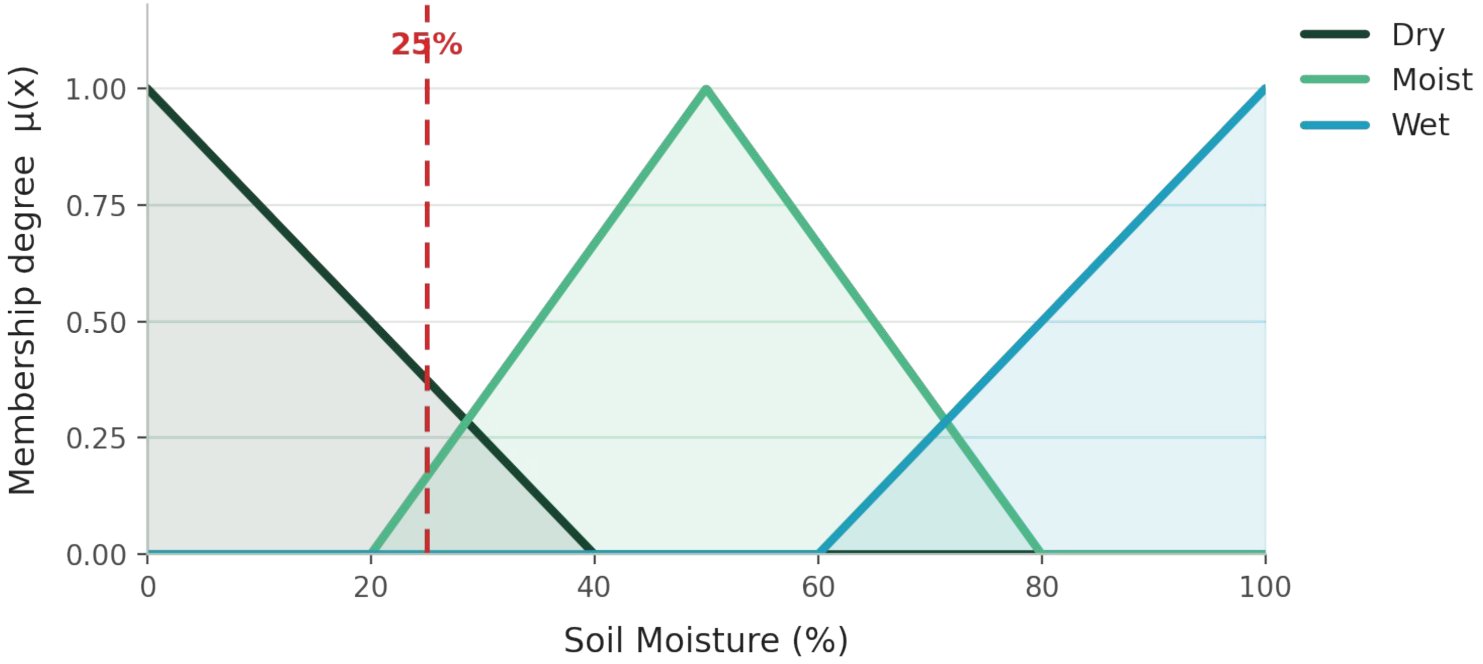
Identify Linguistic Variables & Terms

Variable	Role	Universe of Discourse	Linguistic Terms
Soil Moisture	Input	0 – 100 %	Dry, Moist, Wet
Temperature	Input	0 – 50 °C	Cold, Warm, Hot
Rainfall	Input	0 – 10 mm	None, Light, Heavy
Irrigation Duration	Output	0 – 60 minutes	None, Short, Medium, Long

Why 3–4 linguistic terms per variable?

Enough terms to capture meaningful distinctions (e.g. a merely warm day shouldn't irrigate like a scorching one), but not so many that the rule base explodes. With 3×3×3 input terms there are up to 27 possible rule combinations — manageable for an expert to author and for students to trace by hand.

Membership Functions: Soil Moisture



Membership Functions — Soil Moisture (%)

Dry (triangular)

`a=0, b=0, c=40`

Moist (triangular)

`a=20, b=50, c=80`

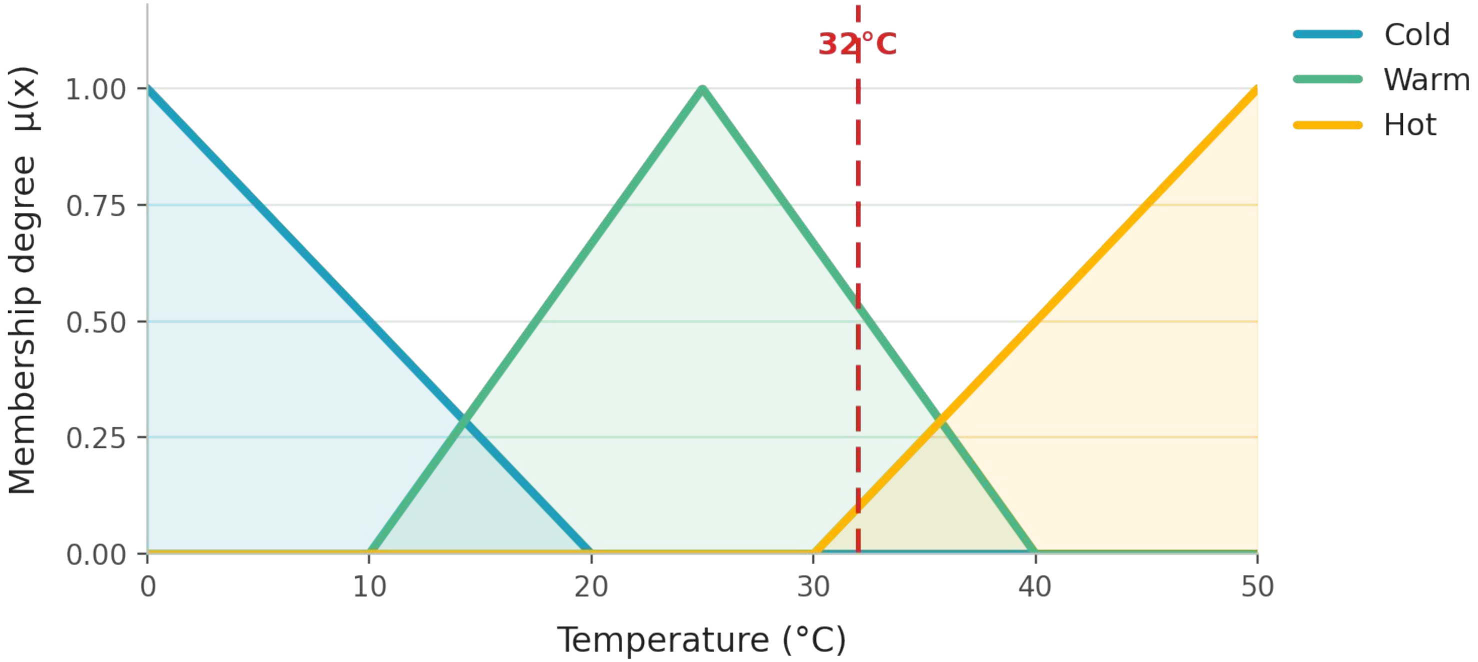
Wet (triangular)

`a=60, b=100, c=100`

Triangular MF formula:
$\mu(x) = \max(\min((x-a)/(b-a), (c-x)/(c-b)), 0)$

Example: a reading of 25% is partially Dry AND partially Moist — both fire simultaneously.

Membership Functions: Temperature



Membership Functions — Temperature (°C)

Cold (triangular)

`a=0, b=0, c=20`

Warm (triangular)

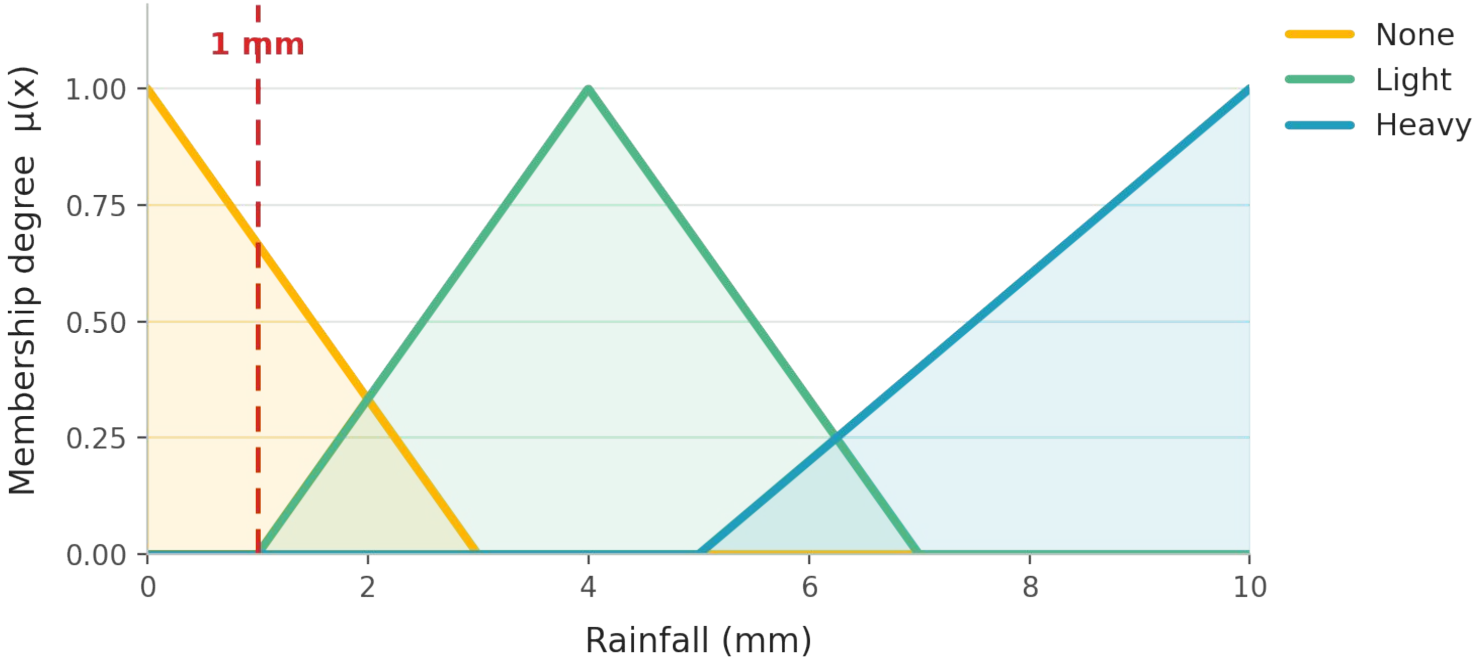
`a=10, b=25, c=40`

Hot (triangular)

`a=30, b=50, c=50`

Shapes are chosen from domain knowledge / agronomic data — not arbitrary. Overlap between adjacent sets is intentional: it gives the smooth transition fuzzy logic is built for.

Membership Functions: Rainfall



Membership Functions — Rainfall (mm)

None (triangular)

`a=0, b=0, c=3`

Light (triangular)

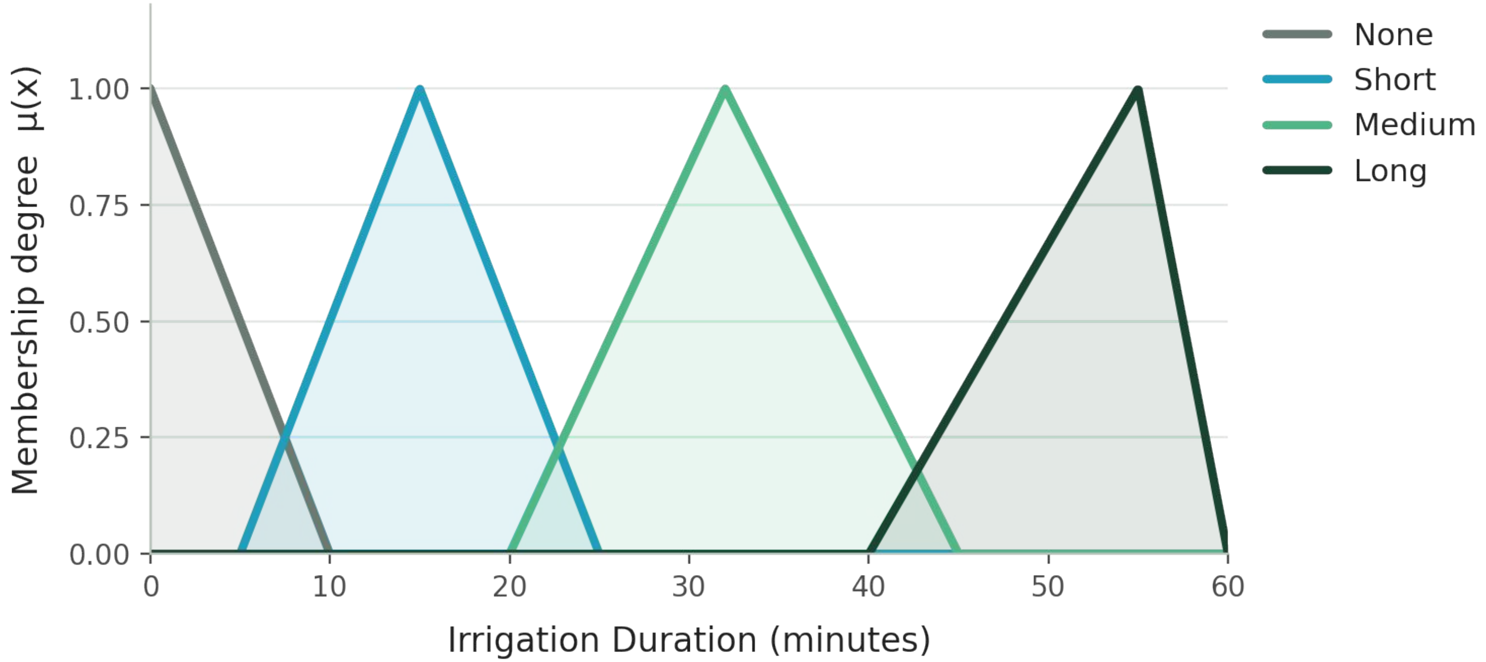
`a=1, b=4, c=7`

Heavy (triangular)

`a=5, b=10, c=10`

Rain acts as an inhibitor: even a dry, hot day needs little irrigation if heavy rain is falling — captured later in the rule base.

Output Variable: Irrigation Duration



Membership Functions — Duration (min)

None (triangular)
`a=0, b=0, c=10`

Short (triangular)
`a=5, b=15, c=25`

Medium (triangular)
`a=20, b=32, c=45`

Long (triangular)
`a=40, b=55, c=60`

The output variable is fuzzified in exactly the same way as the inputs — this is what lets the inference engine and defuzzifier work with it.

Fuzzy IF–THEN Rule Base

Each rule encodes one piece of expert reasoning. AND → take the minimum of antecedent memberships (Mamdani); OR → take the maximum.

#	IF Soil Moisture	AND Temperature	AND Rainfall	THEN Irrigation Duration
R1	Dry	Hot	None	**Long**
R2	Dry	Warm	None	**Medium**
R3	Dry	—	Light	**Short**
R4	Moist	—	None	**Short**
R5	Moist	Hot	—	**Medium**
R6	Wet	—	—	**None**
R7	—	—	Heavy	**None**
R8	Dry	Cold	—	**Short**

Fuzzy Inference: The Mamdani Method

1. Evaluate Antecedents

For each rule, read the membership degree of every input term involved.

```
μ_Dry(25) = 0.375
```

2. Combine with AND / OR

AND → minimum of the memberships. OR → maximum of the memberships.

```
w = min(μ_Dry, μ_Hot, μ_None)
```

3. Apply Implication

The rule's firing strength w clips (truncates) the output term's membership function.

```
μ'_out(y) = min(w, μ_out(y))
```

Why "min" for AND?

A statement joined by AND is only as true as its weakest part — exactly like classical logic's "A ∧ B is true only if both A and B are," generalized to degrees of truth instead of 0/1.

WORKED EXAMPLE — PART 1

Fuzzify a Live Sensor Reading

Sensor snapshot: Soil Moisture = 25% Temperature = 32°C Rainfall = 1 mm

Soil Moisture = 25%

Term	Calculation	μ
Dry (0,0,40)	(40−25)/40	0.375
Moist (20,50,80)	(25−20)/30	0.167
Wet	—	0

Non-zero degrees carry forward into the rule base — zero-membership terms play no further part.

Temperature = 32°C

Term	Calculation	μ
Warm (10,25,40)	(40−32)/15	0.533
Hot (30,50,50)	(32−30)/20	0.100
Cold	—	0

Non-zero degrees carry forward into the rule base — zero-membership terms play no further part.

Rainfall = 1 mm

Term	Calculation	μ
None (0,0,3)	(3−1)/3	0.667
Light (1,4,7)	boundary	0
Heavy	—	0

Non-zero degrees carry forward into the rule base — zero-membership terms play no further part.

WORKED EXAMPLE — PART 2

Evaluate the Rules that Fire

Only rules whose every antecedent has μ > 0 contribute. From the fuzzification step: Dry=0.375, Moist=0.167, Warm=0.533, Hot=0.100, None=0.667 (all other terms = 0).

Rule	Antecedent (AND → min)	Firing Strength w	Output Term Clipped
R1	`min(μ_Dry, μ_Hot, μ_None) = min(0.375, 0.100, 0.667)`	**0.100**	Long
R2	`min(μ_Dry, μ_Warm, μ_None) = min(0.375, 0.533, 0.667)`	**0.375**	Medium
R4	`min(μ_Moist, μ_None) = min(0.167, 0.667)`	**0.167**	Short
R5	`min(μ_Moist, μ_Hot) = min(0.167, 0.100)`	**0.100**	Medium

Note — R2 and R5 both point to “Medium.”

When two or more rules recommend the same output term, their clipped shapes are combined with max (OR) during aggregation — R2’s 0.375 dominates R5’s weaker 0.100 for the Medium set.

Aggregation & Defuzzification — Theory

Step 5 — Aggregation

Every fired rule produces a clipped output membership shape. Aggregation merges all of them — typically with max — into one combined fuzzy output set for the whole system.

$$\mu\_agg(y) = \max(\mu'\_R1(y), \mu'\_R2(y), \ldots, \mu'\_Rn(y))$$

Result: one irregular, possibly multi-peaked shape over the full output universe (0–60 minutes).

Step 6 — Defuzzification

A pump controller cannot act on a fuzzy shape — it needs one number. Defuzzification converts µ_agg(y) into a single crisp value. The Centroid (Center of Gravity) method is the most common Mamdani choice: it balances the “weight” of the whole aggregated shape.

$$y^* = \int y \cdot \mu\_agg(y)\, dy \;/\; \int \mu\_agg(y)\, dy$$

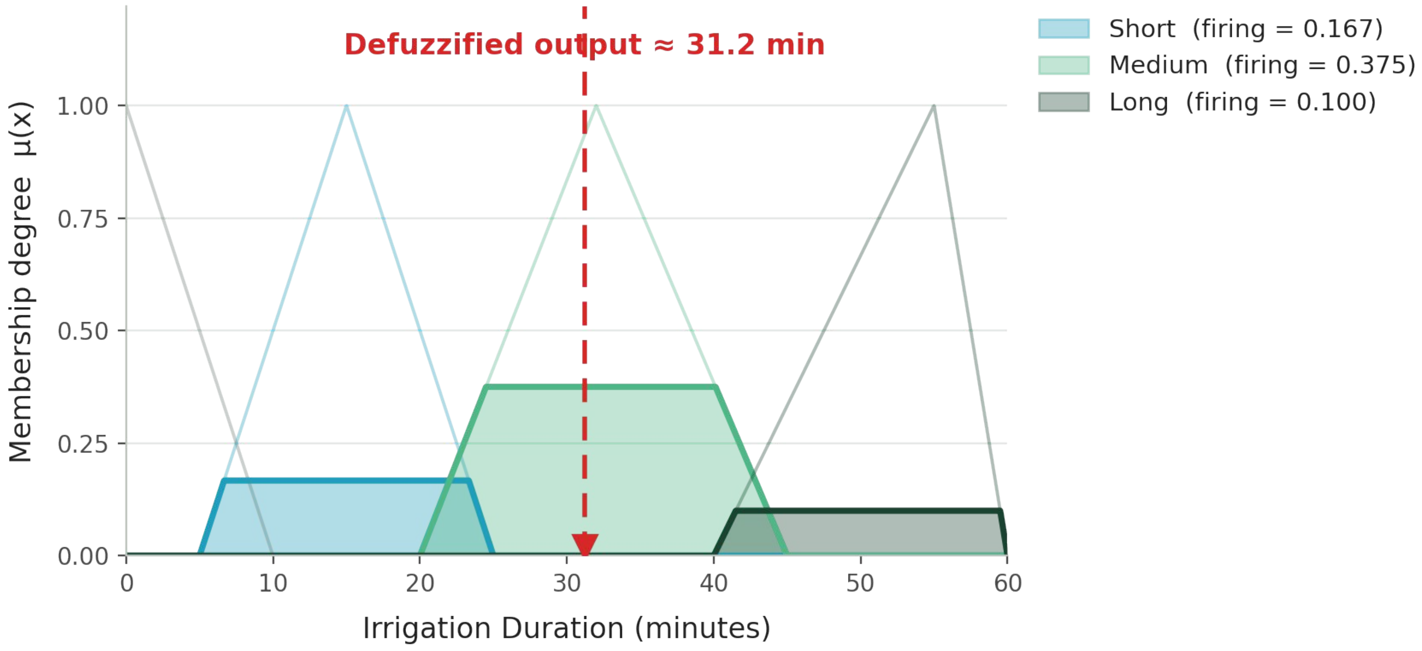
Classroom shortcut — weighted average of each fired term’s peak, weighted by firing strength:

$$y^* \approx \Sigma(w_i \cdot peak_i) \;/\; \Sigma\, w_i$$

(other methods: bisector, mean-of-maximum, smallest/largest-of-maximum)

WORKED EXAMPLE — PART 3

Aggregate the Outputs & Defuzzify



Weighted-Average Defuzzification

Term	w_i	Peak	w_i·peak
Short	0.167	15	2.51
Medium	0.375	32	12.00
Long	0.100	55	5.50

$\Sigma w_i = 0.642$ $\Sigma w_i \cdot \text{peak} = 20.01$

$y^* = 20.01 / 0.642$

≈ 31.2 minutes

Interpretation: moderately dry soil + warm-to-hot air + negligible rain ⟶ the system commits to a moderate ~31-minute watering cycle, not the maximum 60 — exactly the graded response a threshold rule could not produce.

DISCUSSION

Why Fuzzy Logic Suits This Problem

Handles Imprecision

Sensor noise and vague expert language (“fairly dry”) are modeled directly instead of forced into sharp thresholds.

Smooth, Graded Control

Irrigation duration changes continuously with conditions — no abrupt jumps between “no water” and “full blast.”

Transparent Rules

IF–THEN rules are human-readable; a farmer or agronomist can inspect and tune them directly.

Multi-Input Trade-offs

Naturally balances competing signals (hot + dry vs. rain incoming) that a single threshold cannot.

YOUR TURN

PBL Activity — Extend the System

1	**Re-fuzzify by hand**	Given Soil = 55%, Temp = 18°C, Rain = 0 mm, compute all membership degrees and identify which rules fire.
2	**Add a 4th input**	Introduce Wind Speed (calm / breezy / windy) since wind increases evaporation — design its membership functions and 3–5 new rules.
3	**Tune the shapes**	Adjust the breakpoints of “Hot” and “Dry” for a desert climate vs. a temperate one — discuss how outputs shift.
4	**Compare defuzzifiers**	Recompute the worked example using bisector or mean-of-maximum; explain any difference from the centroid result.
5	**Bridge to Genetic Algorithms**	Propose how a GA could automatically tune membership function breakpoints or rule weights instead of hand-tuning them.

SUMMARY

Key Takeaways

1	**Crisp ⟶ Fuzzy**	Real-world variables rarely have sharp boundaries; membership functions model degree of truth.
2	**Fuzzification**	Crisp sensor readings become membership degrees $\mu \in [0,1]$ for each linguistic term.
3	**Rule Base**	Expert IF–THEN rules connect input terms to output terms using AND / OR.
4	**Inference (Mamdani)**	min combines AND antecedents; the result clips (implicates) the output term.
5	**Aggregation**	max merges every fired rule's clipped output into one fuzzy set.
6	**Defuzzification**	Centroid (or weighted average) collapses that set into one crisp control value.

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Further Reading

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